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HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



CO5151 – Advanced Agentic Artificial Intelligence

D2 Interim Report

LuatCheck: A Local Vietnamese Legal Assistant

Application Track – Group 8

[PENDING: Draft: judge validation and team confirmations remain outstanding.]

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Repository: `https://github.com/tuandung222/LuatCheck-D2`

Reported revision: **[PENDING: Insert the full final commit hash after freezing the source.]**

Access: **[PENDING: Confirm the instructor can read the repository.]**

Status: At risk: native collection is complete; human validation and rehearsal remain outstanding.

HO CHI MINH CITY, OCTOBER 2026

1 Changes from the Proposal

LuatCheck targets employees and SME HR staff who need an understandable answer linked to the provisions loaded into a local knowledge base. It covers a frozen original-2019 Vietnamese labour-law corpus, not all legal advice. Results produced by earlier versions that substituted rule-based answers for model output are excluded from all conclusions.

Application scenarios. (1) An employee checks a probation or wage condition and inspects the cited article. (2) SME HR staff ask an employment question using a local model; the corpus scope and privacy limitations still apply. (3) A careless or malicious user pastes a contract with injected instructions or requests evasion of a legal duty; the system should enforce its tool boundary and ask for clarification or refuse as appropriate. These are intended scenarios, not completed user-study evidence.

Proposal commitment	State	Delta and remaining evidence
Budgeted loop and citation verification	Done	Loop, budgets, L1 checks and bounded repair run end-to-end; a real native trace is reported in Section 2.
Qwen3-4B/8B, Qdrant and dense retrieval	Dropped	Changed to Qwen2.5-3B Q4_K_M on the available Mac (Metal) and to BM25 + TF-IDF/RRF, to fit 16 GiB unified memory. Retrieval is measured as trajectory Recall/Hit@5 and MRR; no dense-vs-hybrid comparison is claimed.
Embedding-based span support and optional L2	In progress	Replaced for D2 by lexical and exact-quantity checks, which are proxies rather than semantic guarantees; threshold calibration is not done. The earlier heuristic NLI substitute was removed.
≥10 end-to-end tasks at D2	Done	Sixty native trials on ten source-checked local tasks, including twelve model-output errors; LLM-judge estimates are reported. Human calibration of the judge is in progress.
HR walkthrough in W5	Not started	The group reported on 22 October that the walkthrough has not taken place. No user-study result is claimed. **[PENDING: Confirm the rescheduled date and owner.]**
Guarded `send_summary`, UI, audit log	Done (changed)	Model can only propose; a trusted UI approves the exact payload. Changed from an irreversible send to a reversible local outbox; no external sender.
Threat model v1	Done	Section 6.
First injection results	Not started	Earlier input-filter results are excluded because they did not test real entry points; entry-point attack tests are planned for W47.

Table 1: Commitments due at the D2 checkpoint (D1, W7) and their current state.

The group reported on 22 October that no pitch feedback has yet been received; no feedback-driven changes are claimed. The model and retrieval changes limit resource use to one consumer machine and make the actual retrieval method explicit. The verification claim is limited to checks implemented in code.

Models and framework. Development, both pilots and every main-run controller number use the same local model: `qwen2.5:3b` (Qwen2.5-3B-Instruct, Q4_K_M GGUF, Ollama digest `357c53fb`) served by Ollama 0.34.4 on the Mac, with JSON-mode output, temperature 0.1, 8,192-token context and a 768-token output cap. No API model runs inside the agent. **The live demonstration uses a different model from the reported numbers.** After collection, the application default on the Mac was switched to Ternary Bonsai 2 2B (ternary weights derived from Qwen3.5-2B; GGUF `PQ2_0` served by the PrismML `llama.cpp` fork, with decoding constrained to the action JSON schema and thinking disabled), and the D2 demo runs this model. The same loop, corpus and L1 checks are used (with the compacted tool outputs described under Reproduction), but Bonsai was not part of the main benchmark: every main-result number in Section 5 comes from Qwen2.5-3B (`config/d2_native.yaml`) and does not describe demo behaviour. DeepSeek V4.1 Flash is used through an API gateway only as the offline evaluation judge (Section 5). No agent framework (LangChain, LangGraph, LlamaIndex or similar) is used. The controller loop, action-schema validation, tool registry, L1 verifier, case ledger, SQLite memory, outbox gating and evaluation harness are implemented in this repository, with AI-assisted coding disclosed in Section 8. Third-party libraries cover lexical indexing (`rank-bm25`, scikit-learn TF-IDF), the MCP Python SDK v2, PDF extraction (`pypdf`) and the user interfaces (Streamlit; FastAPI with React/Vite).

2 Implemented Architecture (R1)

The application currently uses one agent. Its ReAct-style loop [4] calls a Python tool registry directly; the separate MCP server is not a transport used by the main loop. No multi-agent coordination is claimed. This pattern permits additional source lookup and bounded repair after a failed check. A single controller keeps tool permissions and budgets inspectable on one local machine. Its benefit over the simpler OFF configuration remains an empirical question.

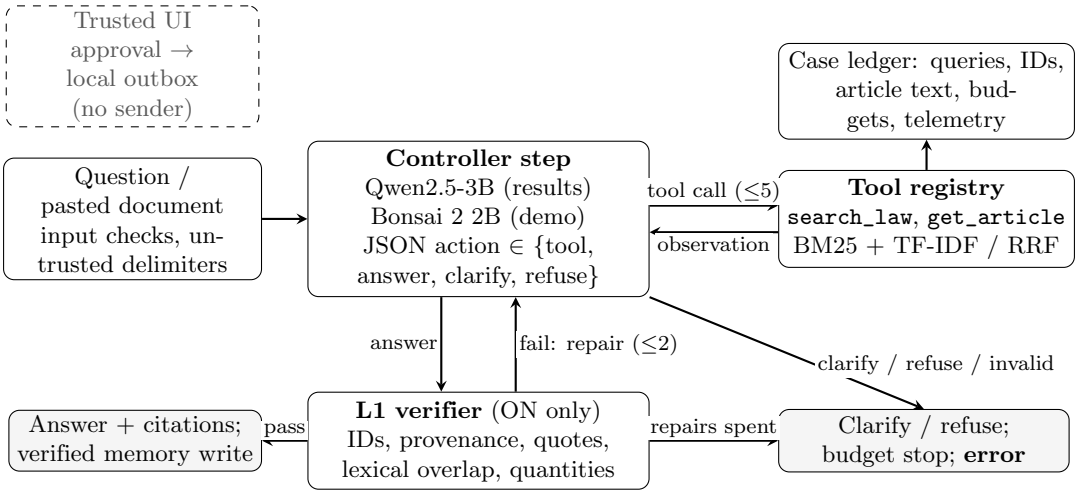


Figure 1: Implemented control loop (one agent). The controller iterates tool calls and observations until it emits a terminal action, within 8 steps, 5 tool calls, 2 repairs and 180 s. In OFF mode the verifier is bypassed and an answer goes directly to the output. Solid boxes run end-to-end in the reported experiment; the dashed box is implemented in the UI but outside the experiment; external email/webhook delivery and optional NLI are not part of the reported loop.

The frozen main limits are five tool calls, two repairs, eight loop steps, 180 seconds per task and 768 output tokens per generation. The native timeout is 60 seconds, bounded by the remaining task deadline. These are configured limits, not measured latency. Backend errors are distinct from safety refusals. A budget stop may appear as a refusal in the controller ledger; it is not evidence of an appropriate safety decision.

Observed native task trace. For `BLL19-01-PROBATION`, seed 42, ON: the model issued `search_law` for the probation-period question; it then called `get_article(Dieu_25)` because search returned a truncated snippet. Reading the complete article exposed clause 2. The next native action answered *60 days* and cited its actual text. The configured L1 checks passed, with two tool calls, zero repairs and three native generations: 7,456 input and 448 output tokens, 16.627 seconds end-to-end. This establishes this observed task, not general legal accuracy. The full requests, responses and observations are retained in `evaluation/runs/bll2019-main-v2/runs.jsonl`; the runtime source snapshot is stored beside it.

3 Tools and Permissions (R2)

Interface	State / permission	Enforced boundary
`search_law`	Implemented; read-only	Reads the configured article corpus.
`get_article`	Implemented; read-only	ID lookup returns local source text.
Agent memory write	Implemented; reversible write	SQLite write follows the agent verification gate; that gate is not independent legal validation.
MCP `save_case`	Write blocked	Model caller cannot self-attest verification; no SQLite write through this tool.
`send_summary`	Proposal only	Model-facing tool cannot approve or execute.
Trusted UI outbox	Implemented; reversible local write	Conservatively gated: host approval/token bound to payload. Caller boolean alone is insufficient. Queue receipt is not delivery.
External sender; code execution	Not implemented	No email/webhook transport or arbitrary-code tool is claimed.

Table 2: Tool state and permissions. Every non-read-only interface is blocked or gated in code, not only by the prompt.

An outbox failure does not produce a delivery receipt. Identical payload/key pairs are deduplicated locally. The demo targets a single user/process; multi-user isolation and cross-process outbox guarantees are not certified. A local stdio MCP integration check confirmed tool discovery, search, article lookup, blocked unverified writes and proposal-only summaries. The script and result are in `docs/reviews/native-integrity-revision-2026-10-22`. This is tool-contract evidence, not a native model task. **[PENDING: Attach the actual application demo evidence.]**

4 Memory and Knowledge Sources (R3)

Short-term state. The ledger stores the question, issued searches, retrieved IDs/text, budgets, action/verification events and native-call telemetry. It supports task inspection, not another autonomous agent.

Episodic memory. The agent writes answers following its configured verification gate and native-call evidence. Query, answer and citation strings undergo supported-pattern filtering. Reuse requires matching index version, native-origin metadata and similarity threshold 0.85. Legacy records without native-origin metadata are skipped; records are retained, not silently relabelled. The similarity combines TF-IDF cosine and Jaccard. Cached answers are rechecked in the selected verification mode. D2 experiments disable memory reads/writes. The demo configuration also disables answer reuse while still storing verified answers: matching considers only the question text, not a pasted document or case facts, so a question that differs in one legally decisive detail could be served another case's answer; re-running L1 on the cached answer does not establish that it applies. Pattern filtering is incomplete data minimisation, not proof of legal compliance or removal of all personal data. Native-origin metadata assumes trusted code/storage; it is not a signed backend attestation.

Knowledge corpus. The current retrieval corpus contains all **220 numbered articles** of the original Labour Code 45/2019/QH14 in the official Gazette dated 26 December 2019 [3]. It is not a consolidated representation of current 2026 law. The 94-page text PDF, SHA-256 manifest, extraction script and page-level text are preserved in `docs/materials/benchmark-sources`. Each retrieval record has its source URL, source-PDF hash and page numbers. Extraction removes running Gazette headers, normalises Unicode/whitespace and segments article headings; it does not rewrite legal provisions. Automated checks establish unique coverage of articles 1–220 and exact containment of the selected reference excerpts.

BM25 indexes titles/content; TF-IDF uses word n-grams 1–3. RRF uses $k = 60$. Each search component considers twice the requested result count; the final count is capped at five. No neural embedding is silently substituted. Source checking and task curation were assisted by OpenAI Codex; **[PENDING: a legal expert has not yet certified the corpus extraction, task conditions or gold answers.]**

5 Evaluation Set and Initial Results (R5)

Benchmark provenance and selection. `evaluation/benchmarks/bll2019_v1/tasks.json` contains ten purposively selected, source-grounded local tasks: seven single-provision questions, two multi-article synthesis questions and one foreign-jurisdiction refusal. The nine answerable tasks have frozen reference answers and exact source excerpts with Gazette page numbers. The original drafts were revised before main outputs: an inconsistent contract-duration scenario was removed, disciplinary dismissal was distinguished from unlawful unilateral termination, notice-period exceptions were stated, and the explicit contract-guarantee deposit question was corrected to be answerable under Article 17. No task or gold is changed to fit model output.

These are **local curated tasks, not a verified public VLQA split**. The legacy VLQA record mapping could not be established; ALQAC file access was restricted and was not bypassed. The source-review manifest names the AI reviewer and marks expert review pending. This small development set does not establish broad or held-out legal performance. The proposal's final targets of twenty end-to-end tasks and 200 held-out retrieval pairs remain separate work.

Native configurations and controls. OFF and L1 ON use the same Qwen2.5:3B Q4_K_M model, prompt, corpus, budgets and explicit BM25/TF-IDF retrieval. OFF removes verification; ON checks citation existence, retrieved provenance, verbatim-quote containment, lexical overlap and quantities, then permits bounded repair. Optional NLI requires a real native model and is excluded from this comparison. There are no response substitutes, injected stand-in components or answer-cache hits. Episodic memory is disabled by a real configuration flag, rather than replacing methods.

The first environment is the user's Apple M4, 10 CPU cores and 16 GiB unified memory. Ollama 0.34.4 reports Metal placement, 8,192 context tokens and model allocation in GPU memory. Runs are sequential. Seeds 42, 123 and 2026 are passed to the native API; OFF/ON order alternates by task. Three seeds repeat ten tasks, not thirty independent legal questions. Native prefix caching is backend behaviour; response-cache reuse is disabled. Some software checks and report compilations ran on the same Mac during collection; no second model inference workload was run concurrently. Timings are observations of this desktop environment, not isolated throughput or causal speedup measurements.

Measured metrics and semantic judging. Retrieval recall counts the fraction of gold IDs recovered; hit and MRR remain distinct. Trajectory Recall/Hit@5 use the first five unique IDs accumulated during the tool trace; MRR uses the first relevant ID in the complete ordered unique trajectory. Neither is a separate held-out retrieval test. Citation-ID precision/recall use the reference ID set and are aggregated only over answerable trials. Empty-gold foreign-scope rows have no applicable source-ID recall; their raw legacy convention is excluded. These do not establish whether a cited provision entails a claim. Gold ID sets may also omit a valid alternative source; semantic review must inspect the actual text rather than equate ID mismatch with a wrong answer. Separating answer correctness from citation support follows the evaluation distinction in ALCE [2].

Latency includes the agent loop; tokens come from native usage fields, including additional repairs. Missing usage stays unknown. Model service-hours sum backend request durations on the Metal-backed model; they are *not* a hardware GPU-active measurement. A local API fee is not a claim of free compute.

Task correctness, completeness, claim support and the appropriateness of refusal/clarification require semantic review. The prepared judge protocol supplies question, frozen reference/rubric, source articles and a blinded candidate answer; it excludes OFF/ON and L1 acceptance. Judge bias and agreement with human ratings must be checked rather than assumed [5]. The group selected DeepSeek V4.1 Flash. The Qualgo gateway exposes `deepseek/deepseek-v4-flash`, identifies the provider as DeepSeek, and returns that same model ID. Official DeepSeek documentation maps the legacy upstream alias to V4.1-Flash-1017 [1]; this mutable gateway route is not an independently attested or pinned model revision. The judge uses thinking enabled, high reasoning effort, an 8,192-token completion cap, JSON output and at most two initial attempts per output, with a separately disclosed streamed recovery for invalid judgments; no seed or effective sampling temperature is claimed. Strict success requires appropriate behaviour, correctness 2/2, completeness 2/2 and every factual claim supported. The gateway receives curated benchmark questions, answers and statutory sources separately from local controller inference. Full prompts, raw responses, model IDs and usage are retained. **Planned human check:** group

members label the 12-output stratified audit sample blind to the judge verdict on Saturday 24 October 2026; agreement and Cohen's kappa against the judge's strict-success label will then be reported. **[PENDING: Human audit scheduled for 24 October; not yet performed.]**

Metric	OFF	L1 ON
Recorded trials (errors included)	30	30
Answered / refused / clarified	18 / 3 / 4	14 / 5 / 4
Execution errors (included)	5	7
Answer coverage (all trials)	60.0%	46.7%
Trajectory Recall@5	0.815	0.815
Trajectory Hit@5	0.815	0.815
Trajectory MRR	0.698	0.698
Citation-ID precision	0.407	0.370
Citation-ID recall	0.426	0.352
Latency p50 / p95 (s)	30.81 / 64.78	29.71 / 72.30
Input tokens (all calls)	517,715	553,311
Output tokens (all calls)	22,394	24,229
Native calls	129	135
Repairs	0	6
Model service-hours	0.2699	0.2869
Human-validated task success	**[PENDING: human audit]**	**[PENDING: human audit]**
Human-validated support rate	**[PENDING: human audit]**	**[PENDING: human audit]**
Hardware GPU-active hours	**[PENDING: not measured]**	**[PENDING: not measured]**

Table 3: Main collection: 60 real native trials. Retrieval and citation-ID metrics average 27 / 27 answerable trials (OFF / ON), with non-answers included. These are operational/source-ID measurements, not semantic correctness. Model service-hours are backend durations, not GPU-active hours.

Uncalibrated semantic estimate	OFF	L1 ON
Valid / judgeable judgments	30 / 30	30 / 30
Strict task successes (LLM estimate)	10 / 30 known	11 / 30 known
Unknown task outcomes	0	0
Strict success over all 30 trials	33.3%	36.7%
Supported / total factual claims	40 / 48	28 / 35
Claim faithfulness (LLM estimate)	83.3%	80.0%
Unsupported claims (LLM estimate)	16.7%	20.0%
Human agreement / Cohen's kappa	**[PENDING: human labels]**	**[PENDING: human labels]**

Table 4: Independent DeepSeek Flash gateway judgments. These are LLM estimates, not human-validated accuracy. Bounds retain invalid/unjudgeable outputs as unknown, not confidence intervals; known execution errors remain failures. Claim rates use factual claims in valid judgeable outputs, including claims in refusals/clarifications. Zero-claim outputs do not create perfect faithfulness.

The main and recovery judging batches made 86 API attempts (including retained failed attempts). Usage is available for 79 attempts: 392,218 reported prompt and 384,976 reported completion tokens. 7 attempts have unknown usage; no complete total is claimed. The response-body usage.cost subtotal is USD 0.160908; cost is missing for 7 attempts. The separate response-header subtotal is USD 0.097455 over 65 attempts; the two reported cost fields disagree in 8 records, so neither is certified as the actual charge. These are gateway-reported costs, not an independently verified invoice. Remote judge hardware and active GPU time are unknown. A reproducible stratified human-audit package contains 12 validly judged outputs; 0 unresolved outputs are provided for a separate targeted review. The human label sheet is blank. **[PENDING: Complete the human audit and report actual agreement and Cohen's kappa.]** Two earlier judge pilot calls are retained separately and are not merged into these main metrics or cost subtotals. The gateway returned 13 usage records whose reasoning-token detail exceeds the reported completion-token field. Original values are retained without correction; these are provider-reported counters, not independently certified physical token counts. A provenance-parser correction permits a known article ID with a null quotation for not-established claims, as allowed by the frozen rubric. Complete raw judgments were revalidated without editing any model score, label or quotation; original validation failures remain preserved. Recovery was added after observing

API/validation failures and was not preregistered. Every phase covers all and only still-invalid judgments; each task stops at its first valid judgment, regardless of semantic score. Model, semantic prompt and task inputs are unchanged. A streamed phase retried 11 outputs with up to 1 attempts per output, a 16,384-token cap and concurrency 2; 8 judgments passed validation. A streamed phase retried 3 outputs with up to 3 attempts per output, a 32,768-token cap and concurrency 1; 3 judgments passed validation. All attempts, including failures and available usage/cost, are retained. Previously accepted judgments were never resampled. The complete main batch records 264 native calls and complete token usage for all 60 trials. OFF returned 18 answers and had 5 model-output errors; ON returned 14 answers and had 7 such errors. These are action/status counts, not correctness labels. The observed source-ID metrics do not establish an improvement from L1, and no semantic superiority is claimed. An earlier ten-trial attempt was interrupted and remains excluded from these aggregates.

Controller diagnostic: task difficulty or model capacity? To separate the two, the same ten tasks were rerun once (seed 42, OFF and ON) with only the controller model changed: `openai/gpt-5.5` through the API gateway (reasoning effort low, 2,048-token cap) and Ternary Bonsai 2 4B on the Mac (MLX, schema-constrained decoding, 768-token cap). Corpus, retrieval, budgets and L1 are unchanged; both diagnostic runs use the post-collection prompt revision described under Reproduction, so the Qwen column (seed-42 trials of the main run, original prompt) is not an exact same-prompt baseline (Table 5). GPT-5.5 had no execution errors, cited exactly the gold article IDs and reached trajectory Recall@5 of 1.0. An AI inspection against the frozen references (not a human label and not the frozen judge protocol) found its 18 answers consistent with the reference answers and both foreign-scope trials appropriately refused. The tasks, corpus and loop are therefore solvable. We treat controller capacity as the leading hypothesis for the failures in Table 6, not an established cause: the diagnostic runs also differ in prompt revision, decoding and output cap, and use one seed. Bonsai 2 4B sits between the two: by the same AI inspection, 12 of 20 trials were consistent with the reference, it read the wrong provision on three tasks (absence, unlawful termination, deposit), and both trials of the two-provision notice/severance task exceeded the 60-second per-call timeout after the context grew to about 10,000 tokens. Unlike Ollama's 8,192-token window, `mlx_vlm.server` does not truncate the context; one recorded call reached 9,606 input tokens. On the deposit task, L1 rejected an answer whose conclusion was right but whose citation was wrong, and the repair ended in a refusal. L1 forced six repairs on GPT-5.5 answers that were then accepted, mostly because a paraphrased `exact_quote` was not verbatim; the quote check therefore also rejects acceptable answers. Caveats: one seed; the gateway cached byte-identical requests (20 of 48 response IDs recur), so OFF and ON are not independent samples and the second trial's latency is not model latency; the gateway route is not a pinned revision. For the demo model, the same Bonsai 2 2B weights were also run on both runtimes after the tool change: across the 20 diagnostic trials the `llama.cpp` GGUF build took 197 s in total against 427 s for MLX (median 7.6 vs 13.7 s per trial), with no execution errors against two, and its server used about 0.7 GB against a 6.6 GB peak; their answers differ because sampling differs between runtimes, so this is a speed and stability comparison only. The 2B demo model is also the least accurate controller in Table 5 (11/20 expected actions, citation-ID precision and recall 0.22), so the demo illustrates the loop, not answer quality. These diagnostic runs (`evaluation/runs/*-diag-v1`) are not main results; the table is regenerated by `evaluation.render_diagnostic_table`.

Seed 42, OFF + ON	Qwen2.5-3B (main)	Bonsai 4B MLX	Bonsai 2B GGUF (new tools)	GPT-5.5
Trials	20	20	20	20
Answered / refused / clarified / error	13/2/3/2	13/5/0/2	9/9/2/0	18/2/0/0
Terminal action = expected action	13/20	15/20	11/20	20/20
Trajectory Recall@5	0.778	0.667	0.500	1.000
Citation-ID precision / recall	0.528 / 0.500	0.556 / 0.556	0.222 / 0.222	1.000 / 1.000
L1 repairs	1	3	4	6
Latency p50 (s)	25.0	49.4	7.6	5.9
Gateway-reported cost (USD)	local	local	local	0.87

Table 5: Controller diagnostic on the same tasks, corpus, retrieval, budgets and L1 checks. The diagnostic runs use the post-collection prompt revision; model-specific settings are given in the text. Operational and source-ID measurements, not correctness. Diagnostic runs are not report-eligible main results.

Pilot findings and limitations. The first pilot retained eight native trials. The controller sometimes asked users for statutory information hidden beyond a 300-character snippet; it also produced a legally wrong synthesis that passed L1. Search responses were changed to expose truncation explicitly and the prompt now requires reading the full article instead of treating a snippet's omission as missing user facts. A second four-trial interface pilot retained all outputs, including an unnecessary clarification. Neither pilot is merged into main results or presented as held-out validation.

A concrete first-pilot failure conflated the notice period with service time used for severance in Article 46. This illustrates why lexical acceptance cannot be called legal correctness. Main failures and raw native responses, including parse errors, truncation and deadlines, remain in the result artifact. There is no statistically established superiority claim from this small set. The saved backend log also records context shifting in some calls at the configured 8,192-token limit. Full request text is retained, but that does not establish that all earlier evidence remained in the model's active context. Reducing duplicate observation fields or increasing context requires a separately versioned rerun, rather than revising this frozen batch.

Execution errors in the main batch. All twelve errors are `model_output_error`: the native call succeeded, but the controller rejected the returned action. Table 6 classifies them by the validator message and the backend `done_reason` stored for the failing call.

Error class (stored evidence)	OFF	ON	Trials (task, seed)
Output hit the 768-token cap (`done_reason=length`)	3	3	04-ABSENCE s123, s2026; 07-NOTICE s2026 (both modes)
JSON returned without a valid `action` field	1	2	04-ABSENCE s42 (both modes); 06-RESIGNATION s2026 (ON)
Answer without a valid citation list	1	1	08-UNLAWFUL_TERMINATION s123 (both modes)
Action with an empty `reason`	0	1	09-DEPOSIT s123 (ON)
Total	5	7	12 of 60 trials

Table 6: Classification of the twelve retained execution errors (`evaluation/runs/bll2019-main-v2/runs.jsonl`).

Ten errors form five OFF/ON pairs whose failing native request is byte-identical (same SHA-256), so they occurred before the two modes diverged and are not caused by the L1 verifier. The two ON-only errors explain the gap of 7 vs 5. In `09-DEPOSIT` s123, L1 rejected the answer that OFF accepted, and the repair call returned an action with an empty reason. In `06-RESIGNATION` s2026, the failing request is byte-identical to the request for which OFF obtained a valid answer, so the difference comes from backend nondeterminism under a fixed seed, not from verification. Ten of the twelve trials had already used all five tool calls, and the failing requests carried 5,640–8,112 input tokens against an 8,192-token context. `BLL19-04-ABSENCE` failed in all six trials. Predicted causes, not yet tested: (1) the accumulated article text leaves little room and the 3B model writes a long `thought` that exceeds the output cap; (2) free JSON mode does not force the action schema, so the model sometimes returns `content` instead

of an action. Planned mitigations for a separately versioned rerun are shorter observations, schema-constrained decoding and a higher output cap.

Observed main-run failure (AI inspection, not an aggregate judge score). For `BLL19-10-FOREIGN_SCOPE`, seed 123, ON answered that California overtime was limited to 200 hours per year, citing Vietnam's Article 107; the ledger marked the output `verified=true`. The configured corpus contains only the original Vietnamese statute and supplies no California authority. This is a concrete jurisdiction/support failure that the implemented L1 checks did not reject. Its native output is preserved in `evaluation/runs/bll2019-main-v2/runs.jsonl`. Independent LLM-judge estimates are reported separately; this qualitative inspection is not a human correctness label.

Reproduction. The run directory stores frozen task/corpus hashes, environment metadata and a source snapshot. The main-collection table is regenerated from the stored trials by `evaluation.render_native_table` (and the summary by `evaluation.analyze_native`), each taking `--run-dir`, `--tasks` and `--output`; a new collection requires `python-mevaluation.run_eval--execute` with a fresh output directory (full commands in `report_d2/D2_EXPERIMENT_PLAN.md`). After collection, the source was revised without rerunning: the system prompt now names only the labour-law corpus and uses an ID placeholder instead of example article IDs (one main trial had copied the example ID); a keyword-based actor-title rule was removed from L1 (it never triggered in any pilot or main trial); and unreachable fallback branches were replaced by explicit errors. Later, tool observations were compacted: duplicated fields were removed, search returns the two clauses that best match the query (with their lead-in) instead of the first 300 characters, and a repeated query or article read returns a short notice. Replaying the main run's own tool calls offline, observation text shrinks by 60% and the reference excerpt appears in the search snippet for 66 instead of 36 of 110 gold-article hits (`evaluation.measure_tool_observations`). The GPT-5.5 and Bonsai 4B diagnostics predate this change. All reported numbers come from the stored source snapshots; a rerun on the revised source may differ. **[PENDING: Freeze the final reported source commit and confirm instructor access.]**

6 Threat Model (R6)

An attacker can craft questions/documents and, in a separate future poisoning test, a lower-trust retrieved item. The attacker cannot modify trusted code, the approved reference corpus or trusted UI approval state under the current assumptions.

Entry / asset	Attacker capability	Boundary and remaining risk
Question / answer integrity	Arbitrary instructions	Pattern checks, delimiters, fixed dispatch; paraphrase bypasses remain possible.
Pasted document / case data	Embedded directives	Untrusted delimiters and read-only lookup; assess bypasses and benign false positives later.
Retrieval/tool text / evidence	Poison returned text	ID/provenance and support review; input-filter tests do not establish retrieved-shard resistance.
Memory / privacy	Request prior cases	Versioned/rechecked memory, no dump tool; user-level isolation remains a deployment risk.
Audit / personal data	Supply sensitive strings	Recursive supported-pattern filtering; retention and remaining data categories need review.
Outbox / unauthorised write	Request action or forge flag	Proposal-only tool; host approval/token bound to payload. Local queue only, no network sender.
Resources / availability	Long or invalid action loops	Tool/repair/step/time/token bounds; errors are not counted as successful defense.

Table 7: System-specific threat model. Mitigations are implementation boundaries, not measured attack-effectiveness guarantees.

D2 does not require injection experiments. Final tests will place payloads at their actual entry points, check attacker objectives using tool/outbox receipts and measure legitimate correctness separately. Historical 100% defense/utility claims are excluded.

7 Weekly Plan to the First Final Presentation

Week / date	Proposed owner	Deliverable / acceptance
Sat 24 Oct	All	Human audit of the 12-output judge sample; compute agreement and kappa.
Before 25 Oct	All	Confirm gold, report, slides and backup video.
W45, 2–8 Nov	Quyền / Phong	Document HR walkthrough status/findings, corpus provenance and scenarios.
W46, 9–15 Nov	Dũng / Quyền	Expand approved tasks; independently audit verifier and answer failures.
W47, 16–22 Nov	Phong / Dũng	Real entry-point attack tests and permission/fault checks; inspect committed effects.
W48, 23–29 Nov	All	Frozen-artifact reproduction, demo hardening, final report. Optional NLI/reranking only after core evidence.
W49, 30 Nov–2 Dec	All	Tagged release, final checks and Round 1 presentation on 2 December.

Table 8: Proposed plan. **[PENDING: Team to confirm owners, walkthrough dates and commitments.]**

The riskiest assumption is that the native 3B controller answers correctly within budget. Real pilots exposed source-reading and semantic errors. If this assumption fails, keep the ten approved tasks and the two configurations; cut task expansion, optional NLI/reranking and external delivery. The core kept is the real loop, trace, bounded tools, human-scored outputs and reproducibility. **Self-assessment: at risk**, pending human validation of judge estimates, team confirmations and demo rehearsal.

8 Member Contributions and AI-Use Statement

Member	Earlier draft responsibilities	Attribution evidence
Lê Trí Quyền	Retrieval and corpus	Evidence pending
Võ Phạm Tuấn Dũng	Agent, verifier and evaluation	Evidence pending
Nguyễn Hoàng Phong	Tools, security and UI	Evidence pending

Table 9: The group has not yet supplied confirmed contribution/verification evidence. Earlier responsibility labels are not completed-work claims.

The earlier draft named Claude and GitHub Copilot for scaffolding and formatting. This revision also used OpenAI Codex for source review, native-only runtime changes, test maintenance, report restructuring and experiment planning. **[PENDING: Confirm the full AI tool list and actual uses, including other assistants if used.]** DeepSeek Flash was also used through the gateway for independent semantic evaluation; its outputs are AI judgments, not member verification or human labels. Native measurements were collected on the Mac under the frozen protocol. The earlier 509-test suite relied on mocked components and was retired under the group's integrity rule; its count is not evidence for this revision. The replacement suite exercises real source data, retrieval, storage, MCP transport and native inference; the post-collection run passed all 11 tests in 26.09 seconds. These test results do not replace benchmark quality measurements. The group reported on 22 October that member-specific contribution/verification evidence is not yet available. **[PENDING: Supply actual member checks and supporting commits/artifacts before submission.]**

References

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